7 Eigenvalues and Eigenvectors in Quantum Mechanics

This chapter introduces the eigenvalue equation, a fundamental concept in quantum mechanics and linear algebra. We explore its mathematical structure, its significance in quantum theory, and its applications to quantum systems and quantum computing. The eigenvalue equation provides the mathematical foundation for understanding measurements, observables, and the behavior of quantum gates.

7.1 The Eigenvalue Problem

The eigenvalue problem is a mathematical concept that arises throughout physics, engineering, and quantum mechanics. It involves finding special vectors, called **eigenvectors**, which, when acted upon by a linear operator (such as a matrix), are only scaled, not rotated or otherwise changed in direction. The scaling factor is called the **eigenvalue**.

Mathematically, the eigenvalue equation is written as:

$$\hat{A} |\psi\rangle = a |\psi\rangle , \tag{7.1}$$

where:

- $\hat{A}$ is a linear operator,
- $|\psi\rangle$ is the eigenvector (or eigenstate),
- a is the eigenvalue, a scalar quantity.

This equation states that when the operator $\hat{A}$ acts on the vector $|\psi\rangle$, the result is a scaled version of $|\psi\rangle$ itself. The eigenvalue a indicates the scaling factor by which the vector is multiplied.

The eigenvalue equation is central to quantum mechanics because it connects the mathematical formalism of the theory with physical observables. In quantum mechanics, operators represent measurable physical quantities (such as energy, momentum, or position), and the eigenvalues of these operators correspond to the possible outcomes of measurements.

7.2 Quantum Operators and Observables

In quantum mechanics, the state of a system is described by a state vector, denoted by $|\psi\rangle$, in a Hilbert space. Physical quantities such as energy, momentum, and position are represented by linear operators acting on these state vectors. These operators are called **observables** because they correspond to quantities that can be measured experimentally.

For physically measurable quantities, the corresponding operators must be Hermitian (self-adjoint), meaning $\hat{A}^\dagger = \hat{A}$. This requirement ensures that the eigenvalues are real numbers, which is necessary since measurement outcomes must be real.

When a quantum operator $\hat{A}$ acts on a state vector $|\psi\rangle$, it generally transforms the state. However, for eigenvectors of the operator, the action of $\hat{A}$ results in simple scaling:

$$\hat{A}\,|\psi\rangle = a\,|\psi\rangle\,. \tag{7.2}$$

The eigenvalues a represent the possible measurement outcomes of the observable associated with operator $\hat{A}$. If a system is in an eigenstate of an observable, measuring that observable will yield the corresponding eigenvalue with certainty.

7.3 Common Quantum Operators

In quantum mechanics, several fundamental operators correspond to physical quantities. Below are commonly encountered quantum operators:

- **Position operator:** $\hat{x}$ (in position representation, this is simply multiplication by x)
- **Momentum operator:**

$$\hat{p} = -i\hbar\frac{\partial}{\partial x} \tag{7.3}$$

- **Energy operator (Hamiltonian):** For time-dependent problems,

$$\hat{H} \text{ is defined via } i\hbar\frac{\partial}{\partial t}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle \tag{7.4}$$

The eigenvalues of these operators correspond to the possible values of position, momentum, and energy that can be observed experimentally. For example, if a wavefunction $\psi(x)$ is an eigenfunction of the momentum operator $\hat{p}$, then a measurement of momentum will yield a definite value—the corresponding eigenvalue.

7.4 Time Evolution and the Schrödinger Equation

The time evolution of a quantum system is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle , \tag{7.5}$$

where $\hat{H}$ is the Hamiltonian operator representing the total energy of the system. This equation describes how the state $|\psi(t)\rangle$ of a quantum system evolves with time.

If the Hamiltonian $\hat{H}$ is time-independent and the system is in an energy eigenstate satisfying $\hat{H} |\psi\rangle = E |\psi\rangle$, the solution to the Schrödinger equation takes the form:

$$|\psi(t)\rangle = e^{-iEt/\hbar} |\psi(0)\rangle . \tag{7.6}$$

Here, E is the energy eigenvalue of the system. This result shows that time evolution for energy eigenstates involves only a global phase factor $e^{-iEt/\hbar}$, which does not affect any measurable probabilities. The probability distributions remain constant in time for energy eigenstates, making them stationary states.

7.4.1 Example: Plane Wave as an Eigenfunction

Consider a free particle described by a plane wave function:

$$\psi(x,t) = Ae^{i(kx-\omega t)}, \tag{7.7}$$

where A is a normalization constant, k is the wave number, and ω is the angular frequency. This function represents a wave propagating in space and time. We now apply the momentum and energy operators to this wavefunction to demonstrate their eigenvalue relationships.

Momentum Operator: The momentum operator in the position representation is:

$$\hat{p} = -i\hbar\frac{\partial}{\partial x}. \tag{7.8}$$

Acting on $\psi(x,t)$:

$$\begin{aligned}
\hat{p}\psi(x,t) &= -i\hbar\frac{\partial}{\partial x}\left(Ae^{i(kx-\omega t)}\right) \\
&= -i\hbar\left(ikAe^{i(kx-\omega t)}\right) \\
&= \hbar k \cdot \psi(x,t).
\end{aligned} \tag{7.9}$$

Thus, $\psi(x,t)$ is an eigenfunction of the momentum operator with eigenvalue $p = \hbar k$.

Energy Operator: For a time-dependent state, the energy operator acts as:

$$i\hbar\frac{\partial}{\partial t}. \tag{7.10}$$

Acting on $\psi(x,t)$:

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\psi(x,t) &= i\hbar\frac{\partial}{\partial t}\left(Ae^{i(kx-\omega t)}\right) \\ &= i\hbar\left(-i\omega Ae^{i(kx-\omega t)}\right) \\ &= \hbar\omega\cdot\psi(x,t). \end{aligned} \tag{7.11}$$

Therefore, $\psi(x,t)$ is also an eigenfunction of the energy operator with eigenvalue $E = \hbar\omega$.

The plane wave solution is simultaneously an eigenfunction of both the momentum and energy operators, with the corresponding eigenvalues being:

$$p = \hbar k, \quad E = \hbar\omega. \tag{7.12}$$

This illustrates the fundamental connection between the mathematical formalism of quantum mechanics and the observable physical properties of quantum systems.

7.5 Quantum States and Quantum Gates

Quantum circuits operate on quantum bits (qubits), the basic units of quantum information. A qubit state can be represented as a vector in a two-dimensional complex Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{7.13}$$

where $\alpha, \beta \in \mathbb{C}$ and the normalization condition requires $|\alpha|^2 + |\beta|^2 = 1$.

Quantum gates act on these state vectors through matrix multiplication. Each quantum gate corresponds to a unitary matrix U (satisfying $U^\dagger U = I$), which ensures that the transformation is reversible and preserves the norm of the quantum state. Common single-qubit gates include:

- **Hadamard gate (H):** Creates superposition states
- **Pauli gates (X, Y, Z):** Perform bit-flips and phase shifts
- **Phase gates (S, T):** Apply specific phase rotations

Multi-qubit gates, such as the CNOT gate, can create entanglement between qubits. A quantum circuit is a sequence of such unitary operations acting on an initial state:

$$|\psi_{\text{final}}\rangle = U_n U_{n-1} \cdots U_1 |\psi_{\text{initial}}\rangle . \tag{7.14}$$

7.6 Measurement and Projection

Measurement in quantum mechanics causes a quantum state to collapse onto one of the eigenstates of the measurement operator. For a qubit measured in the computational basis, the state collapses to either $|0\rangle$ or $|1\rangle$.

Let the state before measurement be $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. According to Born's rule, upon measurement in the computational basis:

- The outcome $|0\rangle$ occurs with probability $P(0) = |\langle 0|\psi|0|\psi\rangle|^2 = |\alpha|^2$,
- The outcome $|1\rangle$ occurs with probability $P(1) = |\langle 1|\psi|1|\psi\rangle|^2 = |\beta|^2$.

These probabilities are obtained by projecting the state vector onto the measurement basis vectors. The squared magnitude of the inner product gives the probability:

$$P(0) = | \langle 0|\psi|0|\psi \rangle |^2 = |\alpha|^2. \tag{7.15}$$

After measurement, the quantum state collapses to the observed eigenstate, and subsequent measurements of the same observable will yield the same result with certainty (assuming no evolution between measurements).

7.7 The Eigenvalue Equation in Quantum Computing

The eigenvalue equation plays a fundamental role in quantum circuits and quantum algorithms. It connects the mathematical formalism with the operational aspects of quantum computation.

7.7.1 Measurement as an Eigenvalue Problem

Every quantum measurement corresponds to a Hermitian operator $\hat{M}$ (an observable), which has a set of orthonormal eigenstates $\{|m_i\rangle\}$ and associated real eigenvalues $\{m_i\}$. When a quantum state $|\psi\rangle$ is measured with respect to this operator, it collapses into one of these eigenstates:

$$\hat{M} |m_i\rangle = m_i |m_i\rangle . \tag{7.16}$$

The outcome of the measurement is the eigenvalue m_i, which occurs with probability $| \langle m_i|\psi|m_i|\psi \rangle |^2$. This provides the quantum mechanical foundation for extracting classical information from a quantum state.

7.7.2 Quantum Gates and Eigenstates

Many quantum gates have well-defined eigenstates and eigenvalues. Understanding these eigenstructures is important for analyzing gate operations:

- The **Pauli-Z gate** has eigenstates $|0\rangle$ and $|1\rangle$ with eigenvalues $+1$ and -1 respectively:

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle. \tag{7.17}$$

- The **Pauli-X gate** has eigenstates $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ with eigenvalues $+1$ and -1.

- The **Hadamard gate** transforms between the Z and X eigenbases, enabling basis transformations in quantum circuits.

Understanding how quantum states align with the eigenstates of gates and measurement operators helps in circuit design and analysis.

7.7.3 Quantum Phase Estimation

In quantum algorithms such as Quantum Phase Estimation (QPE), the eigenvalue equation is central to the algorithm's operation. Given a unitary operator U and one of its eigenstates $|\psi\rangle$:

$$U |\psi\rangle = e^{2\pi i\phi} |\psi\rangle , \tag{7.18}$$

the QPE algorithm estimates the eigenvalue phase ϕ. This capability forms the basis for solving problems in quantum chemistry, number theory (including Shor's algorithm), and optimization.

7.7.4 Stability of Eigenstates

If a quantum system is prepared in an eigenstate of a measurement operator, repeated measurements of that observable yield the same result deterministically. This stability is important in quantum algorithms where consistent outcomes are required. It also plays a role in identifying decoherence-free subspaces in quantum error correction, where certain states remain protected from specific types of noise.

7.8 Summary

The eigenvalue equation $\hat{A} |\psi\rangle = a |\psi\rangle$ is a fundamental mathematical tool in quantum mechanics and quantum computing. It describes how operators act on special vectors (eigenvectors) by scaling them with eigenvalues. In quantum mechanics, this equation connects physical observables with measurement outcomes: the eigenvalues represent possible measurement results, and the eigenstates represent the states in which those results occur